

BABAYEVO, Russian Federation

WMO: 270080

Lat: 59.4042N

Lon: 35.9114E

Elev: 450

StdP: 14.46

Time Zone: 3.00 (E03)

Period: 90-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.2	-11.9	-23.3	1.5	-18.0	-16.8	2.2	-11.5	17.9	27.2	14.7	24.9	1.2	340	0.595

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	82.1	64.9	78.2	63.3	75.1	61.6	67.3	76.8	65.4	75.2	63.6	72.3	5.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.9	90.6	71.0	62.0	84.6	68.7	60.0	78.7	67.0	32.0	77.0	30.5	74.1	29.1	72.5	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.1	13.2	11.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.3	17.5	17.6	26.1	38.4	49.1	59.9	61.9	57.7	48.0	38.3	23.4	20.2
	DBStd	18.87	12.77	13.98	8.56	8.70	8.51	7.52	5.40	5.84	7.57	8.43	12.50	11.72
	HDD50	5369	1009	907	739	361	120	5	1	9	127	370	797	924
	HDD65	9827	1474	1327	1204	799	497	192	125	233	512	828	1247	1389
	CDD50	1096	0	0	0	12	92	302	370	247	66	7	0	0
	CDD65	80	0	0	0	0	5	39	29	6	1	0	0	0
	CDH74	906	0	0	0	2	49	440	310	88	17	0	0	0
	CDH80	187	0	0	0	0	7	118	57	3	2	0	0	0
	CDH80	187	0	0	0	0	7	118	57	3	2	0	0	0
(14) Wind	WSAvg	4.7	5.4	5.8	5.6	4.7	4.7	4.3	4.0	3.7	3.9	4.9	4.5	5.3
(15) Precipitation	PrecAvg	25.4	1.8	1.4	1.3	1.3	2.1	3.0	2.8	2.9	2.3	2.2	2.2	2.0
	PrecMax	30.2	3.2	2.4	2.6	2.7	3.5	5.1	4.8	7.4	5.1	4.3	4.1	3.3
	PrecMin	17.0	0.7	0.2	0.3	0.4	0.6	0.5	0.7	0.4	1.2	0.7	0.1	0.9
	PrecStd	3.3	0.6	0.6	0.7	0.7	0.8	1.2	1.4	1.5	1.0	0.7	0.9	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.6	40.7	49.5	70.7	79.6	87.6	85.8	79.6	76.9	63.3	45.9	38.7
		MCWB	35.4	37.6	41.7	56.8	63.4	65.8	67.2	64.8	63.0	56.3	43.5	37.8
	2%	DB	34.9	37.8	43.4	65.1	73.8	83.5	81.0	76.6	69.7	55.8	42.2	36.7
		MCWB	33.6	36.3	36.7	51.8	57.4	65.7	64.1	61.5	60.0	50.9	40.6	35.7
	5%	DB	33.9	35.7	39.5	59.7	69.4	79.5	77.5	73.6	63.5	52.3	39.9	35.1
		MCWB	32.7	34.1	35.6	48.8	55.9	64.0	62.9	61.1	57.1	48.8	38.9	34.0
	10%	DB	32.8	33.9	37.2	53.3	64.7	75.0	74.1	69.9	59.6	49.5	37.3	33.5
		MCWB	31.8	32.6	34.4	44.8	54.0	62.3	61.6	59.4	54.7	47.2	36.4	32.4
	0.4%	WB	35.5	38.5	42.1	59.6	63.9	71.2	70.2	67.1	65.5	56.4	44.1	38.3
		MCDB	36.0	40.2	49.0	70.5	76.7	81.9	78.2	75.3	72.3	61.2	45.3	38.5
	2%	WB	34.0	36.5	38.4	53.2	60.8	67.7	67.0	64.7	61.1	52.3	41.2	35.8
		MCDB	34.5	37.7	41.2	64.5	70.0	77.2	76.7	71.9	66.7	54.9	42.1	36.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	33.0	34.3	36.4	49.2	57.6	65.7	65.0	62.8	57.9	49.3	38.9	34.1
		MCDB	33.7	35.3	38.6	57.6	66.0	76.5	74.5	69.8	63.0	51.4	39.8	34.7
	10%	WB	31.9	32.7	34.6	45.5	55.0	63.7	63.1	60.9	55.5	47.3	36.6	32.5
		MCDB	32.8	33.8	36.8	52.3	63.6	73.6	71.2	67.7	59.2	49.4	37.4	33.4
	5% DB	MDBR	11.4	13.2	15.0	16.6	18.3	18.8	18.6	16.6	14.1	10.2	9.7	9.8
		MCDBR	7.7	9.0	16.8	28.3	26.7	25.3	25.0	24.5	19.6	14.5	8.0	6.7
(35) Mean Daily Temperature Range	5% WB	MCWBR	7.5	8.4	12.4	17.1	13.3	10.8	11.4	12.3	11.6	10.3	7.5	6.8
		MCDBR	8.4	9.5	12.2	25.7	22.4	21.8	21.5	19.2	18.4	12.1	7.9	7.2
	5% WB	MCWBR	8.5	9.1	9.6	16.6	13.2	10.6	12.4	11.1	11.6	9.4	7.5	7.3
		MCDBR	8.5	9.1	9.6	16.6	13.2	10.6	12.4	11.1	11.6	9.4	7.5	7.3
(40) Clear-Sky Solar Irradiance	taub		0.266	0.307	0.327	0.372	0.363	0.359	0.392	0.385	0.351	0.329	0.294	0.249
	taud		2.076	2.073	2.131	2.228	2.406	2.441	2.359	2.363	2.437	2.391	2.190	2.041
	Ebn at Noon		161	214	252	262	275	278	264	256	247	213	151	122
	Edh at Noon		15	26	34	37	33	33	35	32	25	20	14	11
(44) All-Sky Solar Radiation	RadAvg		111	326	792	1219	1604	1699	1645	1290	759	318	99	58
	RadStd		15	73	108	150	121	110	211	102	84	57	17	10

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	+1.44	N/A	N/A	N/A	N/A	N/A	N/A	-473	-535	N/A	N/A

Nomenclature: See separate page